

Mark G. Scott

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SUMMARY

Data analyst and systems engineer with a background in quantitative social science research. Currently building and automating the operational data systems behind a home energy services contractor while completing an M.S. in Data Analytics and Computational Social Sciences. Comfortable across the full pipeline: research design, data collection and cleaning, modeling, and the systems that keep the data moving.

TECHNICAL SKILLS

Languages R, Python

Methods Machine learning, network analysis, text analysis, spatial analysis, panel regression, research design

Libraries tidyverse, ggplot2, GeoPandas, linearmodels, PyTorch, OpenCV, btergm

Tools HubSpot, SPSS, Zapier, Stata, Git, Linux/WSL2

Systems Workflow automation, API integration, data pipeline design, regulatory compliance systems

EDUCATION

M.S., Data Analytics and Computational Social Sciences

Expected May 2027

University of Massachusetts Amherst, Amherst, MA

GPA 4.00/4.00

Relevant coursework: Machine Learning, Advanced Network Analysis, Advanced Text Analysis, Research Design and Methods, Spatial Analysis

B.A., Political Science

May 2024

Montclair State University, Montclair, NJ

GPA 3.63/4.00

Dean's List 2023–2024 · Certificate of Recognition for Solving Authentic Complex Challenges (2024)

EXPERIENCE

Administrative Assistant → Systems Engineer

August 2025 – Present

Green Collar, South Hadley, MA

- Built an automated scheduling system for a sales team, replacing a manual assignment process.
- Designed a permitting workflow tracking municipal requirements jurisdiction by jurisdiction, and led the effort to bring the company into compliance with local jurisdictions' regulations and licensing requirements.
- Developed an estimation model to score and prioritize prospects, directing sales effort toward higher-value opportunities, and allowing for integration with business in other sectors.
- Administered HubSpot CRM - workflow automation, data integrity, and integrations with external platforms.
- Led an SMS platform evaluation for the CRM: analyzed workflow data, shortlisted three vendors, and built a cost model that drove the selection decision.

Research Assistant

January 2025 – August 2025

APPRISE Inc., Princeton, NJ

- Designed and implemented survey protocols for ongoing research projects.
- Developed methods that increased survey response rates, then standardized them across the research staff.
- Built and maintained a database tracking research activity across concurrent projects.

Research Assistant

January 2024 – May 2024

Montclair State University, Montclair, NJ

- Collected, cleaned, and processed data from online databases, library archives, and expert interviews; produced literature summaries for project leads.
- Coded and labeled data from academic papers, surveys, and interviews for computational analysis.

SELECTED PROJECTS

Economic Dependence and UN Voting Alignment with China

2026

Python · panel econometrics

- Tested whether Chinese trade and investment dependence predicts UN voting alignment using panel fixed-effects regression (linearmodels PanelOLS) with cluster-robust standard errors.
- Adjudicated competing specifications with likelihood-ratio tests, AIC, and Wald tests, documenting why the cluster-robust result is authoritative.

DOJ FOIA Corpus - Redaction Detection and Email Network Analysis

2025 – 2026

Python, R · computer vision, network analysis

- Built Python pipelines to scan roughly 1.38 million DOJ PDFs, extract email headers, and detect redactions with OpenCV and Mask R-CNN (Detectron2).
- Constructed a temporal email network in R: OCR name disambiguation via regex and `refinr`, then yearly network objects prepared for `btergm` modeling.